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NARULA INSTITUTE OF TECHNOLOGY
An Autonomous Institute under MAKAUT
2025

Odd Semester - 2025-26

CS102 - INTRODUCTION TO ARTIFICIAL INTELLIGENCE

Time: 3 Hrs

Maximum Marks : 60

Instructions to the candidate:

Figures to the right indicate full marks.

Draw neat sketches and diagram wherever is necessary.

Candidates are required to give their answers in their own words as far as practicable

Part A

Answer any ten from the following, choosing the correct alternative of each question: 10×1=10

1. An expert system is divided into ____ Subsystems. (1) C02 BL2 PO1
(a) Four (b) Three
(c) Two (d) One
2. What is the field of Natural Language Processing (NLP)? (1) C03 BL3 PO1
(a) Computer Science (b) Artificial Intelligence
(c) Linguistics (d) All of the mentioned
3. What is an AI 'agent'? (1) C02 BL2 PO1
(a) Takes input from the surroundings and uses its intelligence and performs the desired operations
(b) An embedded program controlling line following robot
(c) Perceives its environment through sensors and acting upon that environment through actuators
(d) All of the mentioned
4. Which one from the options would return true/yes for given PROLOG program? (1) C02 BL3 PO3
boy(john,123).
girl(jane,234).
student(john,123).
(a) ?- girl(jane,x). (b) ?- boy(john,123).

- (c) Both a) and b). (d) None of the above
5. Which of the following is an application of Artificial Intelligence? (1) C02 BL2 PO2
- (a) It helps to exploit vulnerabilities to secure the firm (b) Language understanding and problem-solving (Text analytics and NLP)
- (c) Easy to create a website (d) It helps to deploy applications on the cloud
6. Which of the following is/are the composition for AI agents? (1) C02 BL2 PO1
- (a) Program only (b) Architecture only
- (c) Program and Architecture (d) None of the mentioned
7. Which of the following machine requires input from the humans but can interpret the outputs themselves? (1) C01 BL1 PO1
- (a) Actuators (b) Sensor
- (c) Agents (d) AI system
8. Which of the following is not a type of Artificial Intelligence agent? (1) C01 BL1 PO1
- (a) Learning AI agent (b) Goal-based AI agent
- (c) Simple reflex AI agent (d) Utility-based AI agent
9. _____ Number of informed search method are there in Artificial Intelligence. (1) C01 BL1 PO1
- (a) 1 (b) 2
- (c) 3 (d) 4
10. Which of the following produces hypotheses that are easy to read for humans? (1) C01 BL1 PO1
- (a) a) Machine Learning (b) b) ILP
- (c) c) First-order logic (d) d) Propositional logic
11. What is the key difference between supervised and unsupervised learning? (1) C04 BL2 PO2

- (a) Supervised learning requires labeled data, while unsupervised learning does not. (b) Supervised learning predicts labels, while unsupervised learning discovers patterns.
- (c) Supervised learning is used for classification, while unsupervised learning is used for regression. (d) Supervised learning is always more accurate than unsupervised learning.

12. What is the "percept sequence" in the context of an AI agent? (1) C01 BL1 PO1
- (a) A chronological history of every sensory input the agent has ever received. (b) The current state of the environment as perceived by the agent at a single moment in time.
- (c) The specific action the agent chooses to perform next. (d) DBMS

Part B

(Answer any four of the following) 5x4=20

13. How security and safety is being benefitted by Artificial Intelligence (5) C05 BL3 PO2
14. Illustrate the dimensions of Artificial Intelligence. (5) C01 BL2 PO1
15. What are the functionalities of the agent function? Define basic agent program.? (5) C02 BL3 PO1
16. Give the general model of learning agent? What is role of agent program? (5) C01 BL2 PO2
17. Determining Performance Measure(P), Environment(E), Actuator(A) and Sensor(S) for the following AI System:
AI Tutor (5) C02 BL3 PO3
18. How does AI contribute to Industry 4.0? Give one example of an intelligent system. (5) C04 BL3 PO3

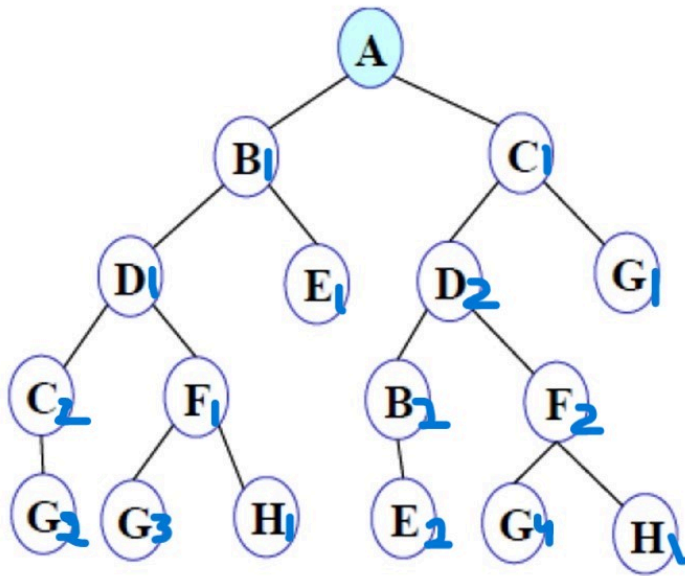
Part C

(Answer any three of the following) 3x10=30

19. (a) Write short note on Breadth First Search in AI. (5) C04 BL3 PO1

- (b) In the figure A is Initial node and F2 is the Goal node, calculate travel path using BFS method.

(5) C04 BL5 PO1



20. (a) Write short note on Compute vision.

(5) C04 BL5 PO2

- (b) Write short note on Reinforcement Learning.

(5) C04 BL5 PO2

21. (a) Write Short notes on Fuzzy Logic and Fuzzy sets.

(5) C03 BL3 PO2

- (b) Write short note on Deep Learning.

(5) C03 BL3 PO2

22. (a) What is Propositional logic in Knowledge representation, explain with example.

(5) C03 BL4 PO1

- (b) Define 9-tiles game as AI game

(5) C05 BL4 PO1

23. Draw a semantic network representing the following knowledge:

(10) C03 BL2 PO1

Every vehicle is a physical object. Every car is a vehicle. Every car has four wheels. Electrical system is a part of car. Battery is a part of electrical system. Pollution system is a part of every vehicle. Vehicle is used in transportation. Swift is a car.